

Architectural Morphogenesis in Autonomous Agent Systems

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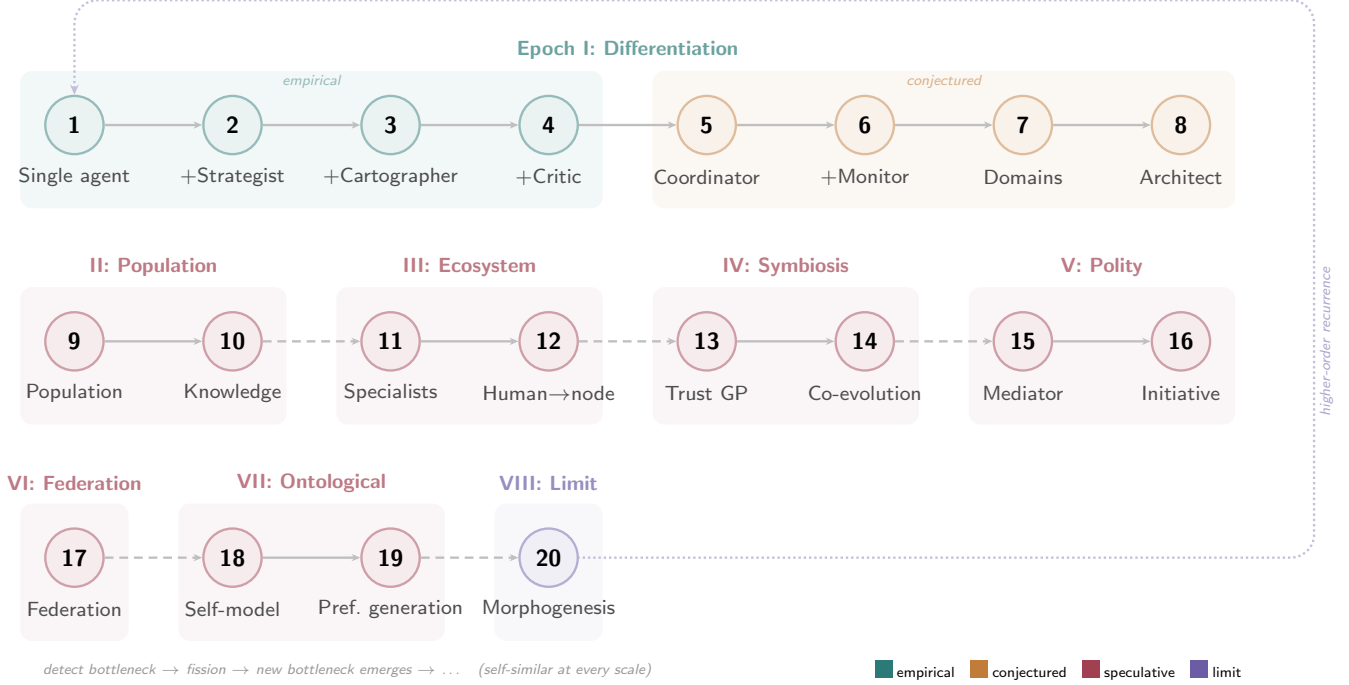


Figure 1: Twenty phases of architectural morphogenesis across eight epochs, graded by epistemic status: observed in the Wallfacer experiment (teal), conjectured from the fission principle (orange), speculatively extended to populations and self-reference (red), and the limiting claim that the stable attractor is morphogenesis itself (purple). Each transition is driven by bottleneck migration: resolving one constraint reliably exposes the next. The dotted loop from Phase 20 back to Phase 1 marks the claim that the entire sequence recurs at higher orders of abstraction.

ABSTRACT

This paper studies architectural change in autonomous agentic software systems through a seven-day longitudinal case study (*Wallfacer*) and a three-run replication experiment from empty repositories (*Ralph*). Across these experiments we report two empirical regularities: *bottleneck migration*, in which relieving one constraint relocates the binding constraint to another system dimension, and convergence-then-divergence, in which independent runs select a similar problem domain while specializing along different developmental trajectories. We use these observations to formulate a testable *role-fission* hypothesis, formalize trust calibration as a Gaussian Process preference-learning problem with uncertainty-aware escalation, and recast architecture selection as a constrained preferential Bayesian optimization. We then place the findings in the literature on LLM-based agents, self-adaptive software systems, cybernetics, and preference learning. The paper is explicit about

epistemic status: empirical results, analytic hypotheses, formal proposals, and speculative extrapolations are separated, and missing evidence is marked with concrete TODOs for future experiments.

KEYWORDS

agentic AI, autonomous systems, preference learning, Bayesian optimization, self-reference, organizational morphogenesis, human-in-the-loop

1 INTRODUCTION

Autonomous agentic AI systems, pipelines of LLM-based agents that plan, execute, document, verify, and sometimes deploy their own work, create a practical version of a classical epistemic problem. When one agent produces code and another agent reviews it, neither judgment supplies an unquestionable ground of correctness. The verification regress, known in philosophy as Agrippa’s trilemma or Münchhausen’s trilemma [1, 51], therefore becomes an engineering

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question: how should an agentic system be structured when no component can be treated as an infallible verifier?

Most current work on LLM agents studies performance under a fixed architecture: a given role decomposition, prompting strategy, tool interface, or benchmark task is evaluated for short-horizon success [31, 39, 45, 60, 61]. Much less is known about *architectural dynamics*: how role structure changes during prolonged autonomous operation, which pressures trigger that change, and whether the resulting dynamics can be modeled rather than narrated post hoc. This paper treats architecture itself as the dependent variable.

We investigate three research questions. **RQ1**: What recurrent patterns of architectural change appear when an autonomous agentic system operates for multiple days without human intervention? **RQ2**: When multiple instances begin from nearly null task specification, which aspects of their developmental trajectory converge and which diverge? **RQ3**: Can the autonomy boundary between agentic systems and human overseers be formalized in a way that supports measurement, learning, and later empirical testing?

To answer these questions, we combine two empirical studies with a formal and conceptual analysis. The *Wallfacer* study is a seven-day longitudinal observation of an autonomous software-engineering pipeline that passed through four architectural configurations. The *Ralph* study is a three-run replication experiment in which independent instances started from empty repositories with no seed task. From these studies we extract two empirical regularities. First, we observe *bottleneck migration*: improvements in one system dimension repeatedly relocate the binding constraint to another. Second, we observe convergence-then-divergence: independent runs converge on the same task family while diverging in implementation language, specialization, and feature strategy.

The paper makes four contributions, ordered by epistemic strength.

Empirical: we document bottleneck migration in Wallfacer and convergence-then-divergence in Ralph. **Analytic**: we propose a *role-fission* hypothesis stating that architectural differentiation becomes likely when conflicting cognitive modes overrun a shared context budget. **Formal**: we model trust calibration as Gaussian Process preference learning with uncertainty-aware escalation and frame architecture search as constrained preferential Bayesian optimization [14, 26]. **Programmatic**: we connect the framework to the literatures on self-adaptive software, organizational design, cybernetics, and security, and we mark missing evidence with explicit TODOs rather than treating speculative claims as established results.

Several scope conditions are important. The empirical base is small: one longitudinal Wallfacer trajectory and three Ralph runs, all on a single LLM substrate and within software-engineering tasks. The architectural transitions in Wallfacer were adaptive interventions, not randomized treatments, so our causal language is intentionally cautious. Accordingly, the paper should be read as a hypothesis-generating systems study plus a formal research agenda, not as a definitive theory of all agentic systems.

The remainder of the paper proceeds from observation to conjecture. Sections 2 and 3 report the Wallfacer and Ralph experiments. Section 4 extracts the role-fission principle and its falsifiable implications. Section 5 formalizes bottleneck migration, trust calibration, and architecture search. Sections 6 and 7 extend the framework into more speculative territory while keeping epistemic status explicit.

Sections 8 and 9 position the work relative to prior literature, enumerate threats to validity and missing experiments, and conclude.

2 THE WALLFACER EXPERIMENT

2.1 Motivation and Setup

The experiment was motivated by a straightforward question: what happens when an LLM-based agent pipeline is left to operate autonomously for an extended period? While single-turn and short-session evaluations of LLM agents are common in the literature, sustained autonomous operation over days or weeks remains underexplored. We sought to observe, rather than prescribe, the architectural pressures that emerge at scale.

The Wallfacer pipeline was deployed on a virtual private server running Ubuntu, with Claude Code [3] as the base LLM substrate. The pipeline was configured for a software engineering task: maintaining and extending a moderately complex codebase. It operated in a self-directed loop encompassing ideation (selecting the next task), implementation (writing code), testing (running the test suite), documentation (updating project records), and deployment (committing and pushing changes). No human reviewed or intervened in any of these stages for the duration of the experiment.

We conducted the experiment across four successive architectural configurations, each introduced when the preceding configuration exhibited sustained performance degradation that we judged, in retrospect, to stem from a structural rather than parametric limitation. The four configurations were: (1) a single agent performing all functions in a self-loop; (2) a Strategist agent separated from an Executor agent; (3) the addition of a Cartographer agent; and (4) the further addition of a Critic agent.¹ Each configuration was run until we observed a clear and persistent bottleneck, at which point we introduced the next configuration. The total duration across all configurations was seven days, during which the codebase grew from approximately 10,000 to over 40,000 lines of code.

Methodologically, this is an adaptive observational study rather than a randomized experiment. Architectural transitions were chosen online in response to observed failure modes, so the section provides evidence about one realized trajectory, not clean causal estimates of what each added role would have done under controlled counterfactual conditions.

TODO: Add a reproducibility appendix with exact prompts, model/version identifiers, token and time budgets, restart policy, logging schema, and the operational rule used to classify a bottleneck as “clear and persistent.” Without these details, the Wallfacer study remains difficult to reproduce.

2.2 Observations and Analysis

We report three principal observations from the experiment, along with our interpretation of each. Because the study is single-case and adaptive, these observations are best read as hypothesis-generating evidence rather than statistically established regularities.

¹We name these roles after their cognitive mode rather than their mechanical function. The Strategist decides *what* to do (not merely “proposes”), the Executor implements (not merely “builds”), the Cartographer produces legible abstractions of system state (not merely “documents”), and the Critic exercises evaluative judgment (not merely “verifies”). These names reflect the fission principle’s emphasis on competing cognitive modes.

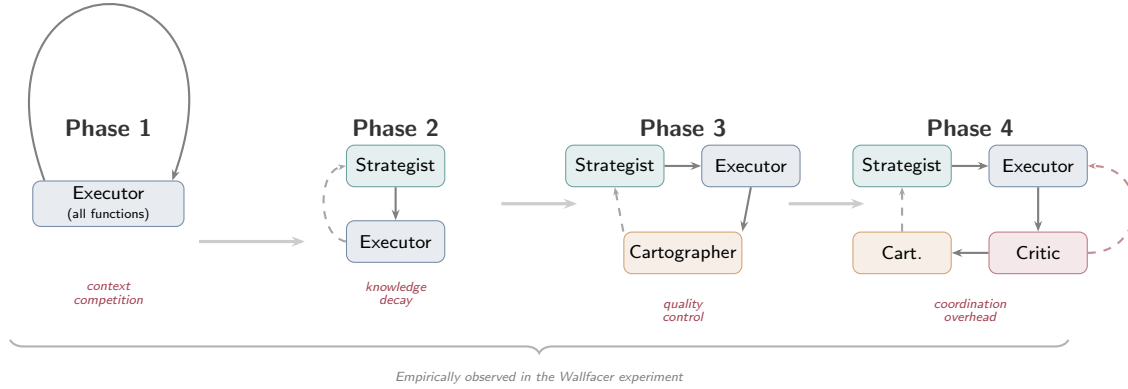


Figure 2: Architectural evolution across four observed phases. Each transition was triggered by a persistent bottleneck (shown in red). Solid arrows indicate primary information flow; dashed arrows indicate feedback.

Observation 1: Bottleneck migration. Across all four configurations, resolving a performance bottleneck in one dimension of system behavior reliably relocated the binding constraint to a different dimension, rather than producing an overall improvement that persisted. For example, in the two-agent configuration, separating planning from implementation eliminated the context-competition bottleneck observed in the single-agent configuration, but introduced a new bottleneck: implicit knowledge generated during implementation (such as the rationale behind design decisions) was not captured in any durable form and therefore decayed between planning cycles. The Cartographer was introduced to address this decay, which it did, but at the cost of introducing a quality-control bottleneck: three agents were producing outputs, but none was tasked with evaluating whether those outputs met any standard of correctness.

This pattern, in which resolving one constraint exposes another, aligns with Goldratt’s Theory of Constraints [25], which posits that any system has exactly one binding constraint at a time and that improving non-constraints yields no net throughput gain. It also resonates with the No Free Lunch theorems [59], which establish that no single optimization strategy dominates across all problem instances. In our setting, the analogous claim is that no single architectural configuration dominates across all bottleneck types. We note, however, that our evidence for this claim is limited to a single system operating on a single class of tasks, and that broader empirical validation is required before any general principle can be asserted with confidence.

TODO: Add round-level measurements that show the targeted dimension improving while a different dimension becomes binding after each architectural transition. A pre/post table or time-series analysis would make Observation 1 materially stronger.

Observation 2: Emergent self-diagnosis. In the three-agent configuration (Strategist, Executor, Cartographer), the system proposed refactoring its own monolithic codebase at approximately 51,000 lines of code. This behavior did not emerge in the two-agent configuration at a comparable scale. We interpret this as evidence that the Cartographer role, by producing structured representations

	Ph.1	Ph.2	Ph.3
Context	binding		
Knowl.		binding	
Quality			binding
Coord.			

Figure 3: Bottleneck migration across phases. Dark cells mark the binding constraint; light cells indicate resolved dimensions. Dashed arrows show how each fission relocates the bottleneck.

of the system’s architecture and history, provided the Strategist with a legible abstraction of global system state that was otherwise unavailable. In the two-agent configuration, the Strategist’s decisions were conditioned only on the current codebase and task queue, without any higher-level summary of architectural trends.

This interpretation finds support in Clark and Chalmers’ Extended Mind hypothesis [15], which argues that cognitive processes are not confined to the brain but extend to external artifacts that participate in reasoning. In our case, the documentation artifacts produced by the Cartographer function as external cognitive scaffolding for the Strategist: they do not merely record what has happened but actively shape what the Strategist can perceive and reason about. We are cautious, however, about overinterpreting a single instance of emergent behavior. It is possible that the refactoring proposal would have emerged eventually in the two-agent configuration at a higher line count, or that it was triggered by idiosyncratic features of the codebase rather than by the architectural difference we hypothesize.

Observation 3: Coordination cost scaling. As the number of agent roles increased from one to four, we observed a nonlinear increase in the volume of inter-agent communication required to maintain coherent system behavior. The four-agent configuration exhibited episodes of circular dependency, in which the Critic rejected output that the Executor had produced according to the Strategist’s specification, leading to repeated revision cycles that

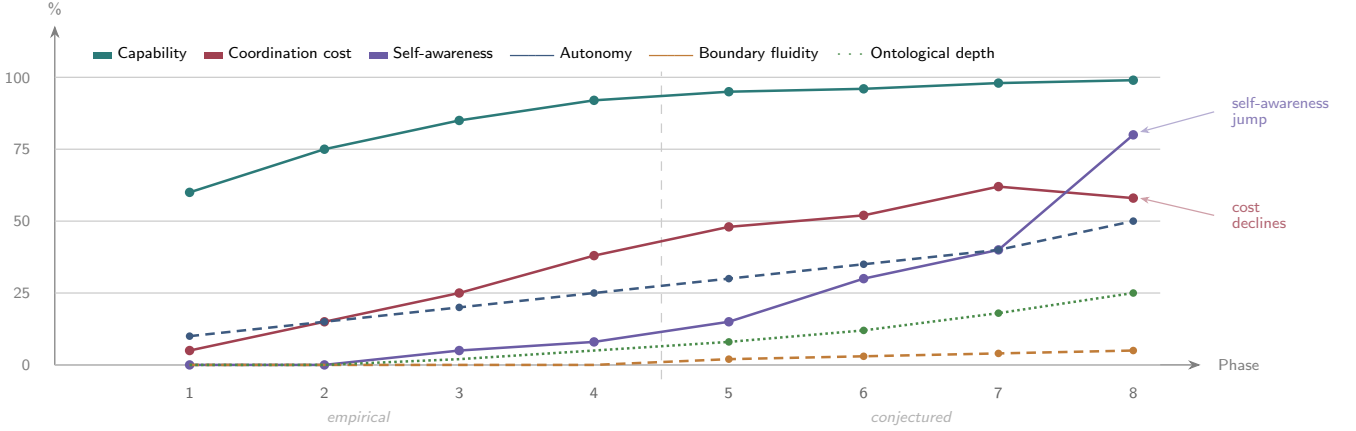


Figure 4: System metric evolution across eight phases. Solid lines represent primary metrics (capability, coordination cost, self-awareness); dashed and dotted lines represent emergent metrics that become significant only in later phases (autonomy, boundary fluidity, ontological depth). Phases 1 to 4 are empirically grounded; Phases 5 to 8 are conjectured extrapolations. The sharp rise in self-awareness at Phase 8 reflects the Architect role’s capacity for architectural self-modification. Values are normalized estimates based on the Wallfacer observations (Phases 1–4) and theoretical projections.

consumed computational resources without advancing the project. This observation is consistent with Brooks’ [11] well-known finding that communication overhead in human teams scales quadratically with team size, and it suggests that the benefits of role differentiation are subject to diminishing returns as coordination costs accumulate. Figure 4 traces the estimated trajectory of six system metrics across all eight phases, illustrating how capability saturates while coordination cost, self-awareness, and autonomy follow distinct growth curves.

TODO: Report communication volume, revision-loop frequency, and test-pass latency per phase. Coordination cost is central to the paper’s explanation but is currently inferred more strongly than it is directly measured.

3 THE RALPH EXPERIMENT: CONVERGENT EVOLUTION WITHOUT SPECIFICATION

The Wallfacer experiment demonstrates bottleneck migration and role fission within a single pipeline. A natural follow-up question is whether multiple independent instances exhibit convergent or divergent evolutionary trajectories, and what factors govern the balance between the two. To investigate this, we conducted a second experiment using a simpler pipeline that lacks Wallfacer’s role differentiation.

3.1 Setup

We used *ralph*, an autonomous agent pipeline that runs a three-phase loop: a **Thinker** analyzes the current project state and proposes exactly one goal for the next round; a **Worker** implements that goal; and a **Committer** stages all changes and pushes a commit. Unlike Wallfacer’s multi-role architecture, *ralph*’s loop is fixed: no Cartographer, no Critic, no fission. The pipeline persists execution history as JSON logs, enabling the Thinker to condition each new goal on the full trajectory of prior rounds.

We pointed *ralph* at three empty Git repositories with no seed task and no specification of what to build. All three instances used the same LLM substrate (Claude) and the same three-phase loop structure. Two instances (Sandbox and Simulator) used the stock Thinker prompt, which provides no strategic guidance beyond a generic instruction to “be creative.” The third instance (Explorer) used a modified Thinker prompt that included two additional signals: awareness of its current round number, and an *explore/exploit score*, a deterministic sinusoidal function of the round number with period ≈ 6 rounds and ± 0.2 random jitter, presented to the Thinker as a percentage (e.g., *explore*=73%) with one of three steering phrases (“lean toward exploration,” “lean toward exploitation,” or “balance”). The signal is not adaptive or learned; it is a fixed oscillation that periodically nudges the Thinker toward deepening existing capabilities rather than adding new ones. This 2+1 design was not planned as a controlled experiment; the prompt modification preceded the experiment and the divergent trajectories were observed post hoc. Nevertheless, the design affords a natural comparison: the two stock-prompt instances serve as near-replications of each other, while the modified instance isolates the effect of a single prompt-level variable. Two instances completed 42 rounds and one completed 33 rounds before being stopped. Table 1 summarizes the resulting codebases.

The resulting evidence is stronger than a single anecdote but still far from a fully powered experiment. Three runs are enough to surface an interesting pattern; they are not enough to cleanly separate prompt effects, sampling variance, and latent substrate priors.

TODO: Add additional seeds per condition, publish the exact Thinker prompts, and report temperature or sampling settings. The current 2+1 design is useful as a pilot study, but stronger claims about the *explore/exploit* signal require replication.

Table 1: Three codebases generated by ralph from empty repositories with no seed task. All three independently converged on cellular automaton simulators.

Repository	Lang.	LOC	Funcs	Cost
Explorer	C	10,876	133	\$89.66
Sandbox	Python	14,727	509	\$56.87
Simulator	Python	12,336	285	\$57.80

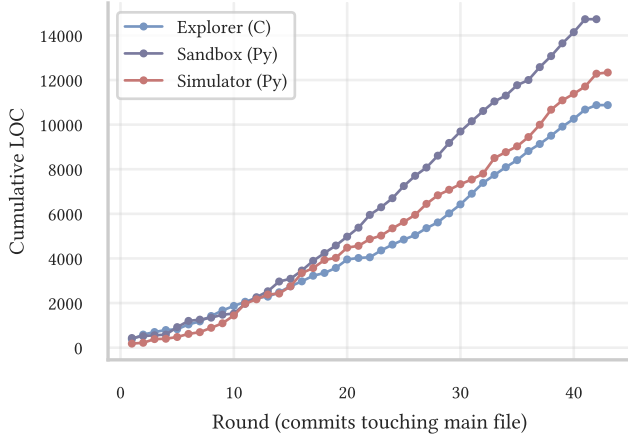


Figure 5: Cumulative lines of code across three repositories initialized from empty directories with no seed task. All three converged on cellular automaton implementations. Growth is monotonically increasing with near-zero churn (3.9%, 0.5%, and 3.0% deletion ratios respectively).

3.2 Observations

Observation 4: Convergent goal selection from null initial conditions. All three instances independently chose to build a cellular automaton simulator as their first goal and continued developing it for every subsequent round. This convergence occurred without any seed task, without inter-instance communication, and despite the prompt-level difference between the Explorer and the other two instances: neither the explore/exploit signal nor its absence influenced the initial domain choice. The “null” initial condition is, of course, not truly null: it includes the LLM’s entire training distribution, which implicitly encodes priors over what constitutes an interesting, tractable, and self-contained software project. The convergence on cellular automata suggests that this class of programs occupies a strong attractor basin in the LLM’s prior over generative tasks, likely because cellular automata are well-represented in training data, produce visually compelling output, and admit incremental extension without architectural refactoring. This parallels convergent evolution in biology, where independent lineages arrive at similar solutions (e.g., camera eyes evolving independently over 40 times [36]) when facing analogous selection pressures.

Observation 5: Divergent specialization along orthogonal axes. Despite convergent initial goals, the three codebases specialized in qualitatively different directions (Figure 6). Explorer, written in C, developed deep scientific analysis capabilities: 15+

information-theoretic overlays (Shannon entropy, Lyapunov exponents, transfer entropy, Wolfram classification), multi-rule zones, wormhole portals, and exotic topologies (torus, Klein bottle, Möbius strip). Sandbox pursued breadth, accumulating 27 interactive simulation modes spanning five categories (cellular automata, physics, biology, procedural generation, algorithms). Simulator combined algorithmic diversity with tooling: 28+ simulation modes alongside genetic algorithm rule discovery, headless batch rendering, and GIF/PNG export.

Figure 7 quantifies this divergence. The Explorer’s mean cyclo-matic complexity (22.9) is over three times that of the Sandbox (6.8), reflecting both the inherent verbosity of C and the depth of its analysis routines. The Simulator accumulated the most simulation modes (131) despite having the fewest rounds (33), suggesting a strategy of adding many small, self-contained variants rather than deepening existing ones.

The 2+1 experimental design allows us to partially disentangle stochastic from prompt-driven divergence. The two stock-prompt instances (Sandbox and Simulator) followed near-parallel trajectories: both defaulted to a breadth-first strategy of adding a new simulation mode each round, and they share 15 of their modes (including Lenia, Wa-Tor, Boids, Lattice Boltzmann, Ising model, and Wave Function Collapse). The Explorer, whose Thinker was periodically nudged toward exploitation, followed a qualitatively different depth-first trajectory: it stayed with one simulation model (Game of Life) and layered increasingly sophisticated analysis on top of it (Shannon entropy, Lyapunov maps, Fourier spectra, topological Betti numbers, renormalization group flow). The near-replication between Sandbox and Simulator, contrasted with the Explorer’s divergence, suggests that the explore/exploit signal, a single scalar variable in the Thinker prompt, is the primary driver of the depth-versus-breadth split, rather than stochastic variation in LLM sampling.

TODO: Test this attribution with a balanced factorial study: multiple seeds, multiple model backends, and at least two deterministic explore/exploit schedules. The current evidence is consistent with a prompt effect, but not yet decisive.

This finding has two implications. First, a minimal prompt-level intervention can redirect an autonomous system’s entire developmental trajectory, suggesting that the explore/exploit balance is a potent control variable for agentic morphogenesis. Second, the near-identical trajectories of the two stock-prompt instances provide evidence that the LLM’s implicit prior over generative tasks is not merely an attractor for initial goal selection (Observation 4) but continues to constrain the full developmental trajectory when no countervailing signal is applied. The pattern of convergence-then-divergence is consistent with adaptive radiation in evolutionary biology [49]: a population that enters a new ecological zone (here, the “cellular automaton niche”) initially exploits the same resources before diverging into specialized sub-niches, but the degree of divergence is modulated by a selection pressure (the explore/exploit signal) that has no direct biological analogue.

Observation 6: Monotonic growth without self-initiated refactoring. Figure 5 shows that all three codebases grew monotonically, with deletion ratios below 4%. None of the three instances ever proposed refactoring its own codebase, in contrast with the

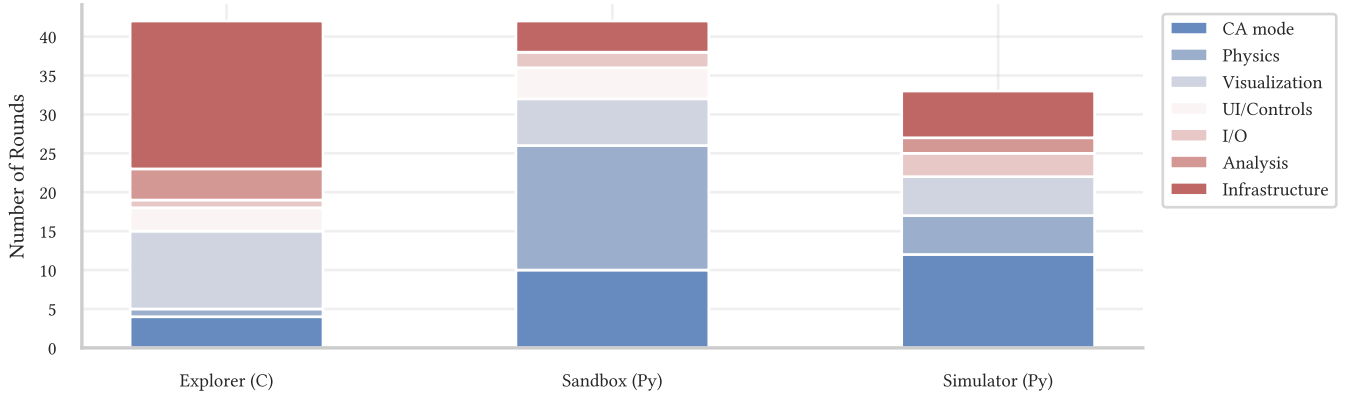


Figure 6: Feature categories per repository, classified by keyword matching on each round’s Thinker goal. Despite convergent initial goals, the three codebases diverged along orthogonal axes: Explorer (modified prompt with explore/exploit signal) toward scientific analysis overlays, Sandbox and Simulator (stock prompt) toward breadth in simulation modes, with Sandbox emphasizing physics and Simulator emphasizing I/O capabilities.

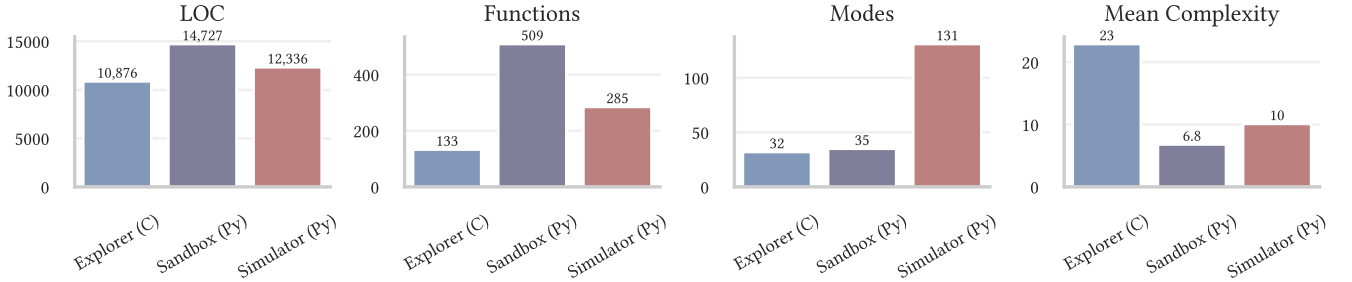


Figure 7: Final codebase comparison across four metrics. The Explorer’s high mean cyclomatic complexity (22.9 vs. 6.8 and 10.1) reflects its C implementation and deeply nested analysis routines. The Simulator’s mode count (131) dwarfs the others, reflecting its strategy of adding many small simulation variants.

Wallfacer experiment’s Observation 2, where the three-agent configuration proposed refactoring at approximately 51,000 lines. We interpret this as negative evidence supporting the fission principle: ralph’s fixed thinker-worker-committer loop lacks a Cartographer role that would produce the structured self-representation needed for architectural self-awareness. Without a legible abstraction of global system state, the Thinker’s planning horizon remains local: it can propose the next feature but cannot diagnose structural debt. This suggests that self-diagnostic capability is an emergent property of role differentiation, not an intrinsic capability of the underlying LLM.

TODO: Compare ralph against an otherwise identical variant with an explicit summary-producing role or persistent architecture report. That ablation would test whether the missing Cartographer, rather than codebase size alone, explains the absence of refactoring proposals.

4 THE FISSION PRINCIPLE

The observations reported above suggest a regularity that we formalize as a *hypothesis* rather than a law.

Principle 1 (Role Fission). *Role fission occurs when a single agent is required to simultaneously serve two cognitive modes that compete for context window resources. The fission event separates these modes into distinct agents, each with a dedicated context allocation and a narrower scope of responsibility.*

The principle should be read as a compact explanatory hypothesis: it predicts when specialization becomes attractive, but it does not claim that fission is always optimal or that only one decomposition is available.

The three fission events observed in the Wallfacer experiment correspond to three specific mode conflicts, summarized in Table 2.

Table 2: Observed fission events and their triggering mode conflicts.

Mode Conflict	Fission Product	Ph.
Planning vs. Implementation	Strategist	2
Execution vs. Reflection	Cartographer	3
Generation vs. Judgment	Critic	4

The Ralph experiment (Section 3) provides negative evidence for this principle: ralph’s fixed thinker-worker-committer loop never underwent fission, and correspondingly, none of the three instances ever proposed refactoring its own codebase despite reaching 10,000–15,000 lines. The self-diagnostic capability that emerged in Wallfacer’s three-agent configuration (Observation 2) was absent in ralph’s undifferentiated pipeline, consistent with the claim that such capabilities are emergent products of role separation rather than intrinsic properties of the LLM substrate.

This pattern bears a structural resemblance to organizational fission in human enterprises. A startup begins with a single individual performing all functions; as the enterprise grows, product management separates from engineering, technical writing separates from development, and quality assurance separates from production. Conway’s Law [17] observes that the structure of a software system tends to mirror the communication structure of the organization that produces it. In the Wallfacer experiment, we observe a form of reverse Conway’s Law: it is not organizational structure that constrains system architecture, but rather capability pressure within the system that drives structural differentiation. We note that Malone and Crowston’s [40] coordination theory provides a formal account of how dependencies between activities give rise to coordination mechanisms, and Simon’s [53] theory of near-decomposable systems explains why decomposition along the axis of weakest coupling tends to produce stable hierarchies.

4.1 Falsifiable Implications

The role-fission principle is useful only insofar as it generates predictions that could fail. Three implications follow directly from the Wallfacer and Ralph studies. First, increasing available context budget or reducing interference through better summaries should postpone the first beneficial fission event. Second, introducing a summary-producing role into an otherwise fixed pipeline should increase the probability of self-diagnostic actions such as refactoring proposals. Third, as role count increases under fixed communication bandwidth, net performance should eventually exhibit diminishing or negative returns.

TODO: Turn these implications into explicit hypotheses with pre-registered outcome measures. For example: probability of refactoring proposals, throughput per token-dollar, defect escape rate, and communication volume per completed task.

4.2 Theoretical Extension: Phases 5 to 8

Beyond the four phases directly observed in the experiment, we conjecture that continued operation would produce additional fission events driven by new categories of bottleneck. We present these as theoretical predictions, not as empirical findings.

Phase 5: Coordinator. The four-agent system’s coordination overhead (Observation 3) would eventually require a dedicated *Coordinator* role to manage global tradeoffs, for instance, when to refactor versus when to add features. No existing agent in Phase 4 is tasked with cross-cutting prioritization; the Strategist proposes tasks and the Critic evaluates outputs, but neither maintains a coherent model of system-wide resource allocation.

Phase 6: Monitor. The Coordinator’s context window and a human manager’s span of control are structurally the same problem:

a single decision-maker overwhelmed by raw signal. We hypothesize that a *Monitor* role would fission from the Coordinator to perform signal compression: continuously observing all agent outputs and condensing them into digestible health indicators. The Coordinator thereby transitions from direct observer to dashboard-based decision-maker. This addresses the span-of-control problem identified by Gulick [27] and implements a version of Conant and Ashby’s [16] Good Regulator theorem: the Monitor’s compressed representation constitutes a model of the system it regulates.

Phase 7: Domain decomposition. A single pipeline’s throughput eventually saturates. The system would decompose by domain into parallel sub-pipelines, each with its own complete role stack. This is Conway’s Law [17] in its agentic form: the system’s architecture begins to mirror its organizational structure. The new binding constraint becomes cross-domain consistency: Pipeline A modifies an API that Pipeline B depends on, but Pipeline B has no mechanism to detect the change.

Phase 8: Self-reconfiguration. This is where, we conjecture, the verification regress grounds out. An *Architect* agent treats the system’s own architecture as an optimization variable. Crucially, the Architect does not require verification by another agent; it calibrates itself through the long-term consequences of its own decisions. Different architectural configurations are compared via pairwise preference judgments, which is precisely the Preferential Bayesian Optimization framework developed in Section 4. Architecture ceases to be a constant and becomes a variable under active optimization.

This last conjecture is the bridge to the formal framework developed in the next section. Whether these specific phases would materialize in practice, and in this particular order, is an empirical question that we cannot answer from the present case study.

TODO: Add a clear stopping rule for speculation. One option is to state which observations would count against the Phase 5–8 ordering, rather than presenting the sequence only as a plausible extrapolation.

5 FORMALIZATION

The aim of this section is not to claim a complete theory, but to convert the paper’s central constructs into measurable objects. A more scientific account of agentic morphogenesis requires definitions that support logging, comparison across runs, and eventual hypothesis testing.

5.1 System State, Outcomes, and Bottleneck Migration

We model an agentic system at round t by a state

$$S_t = (A_t, W_t, M_t, H_t), \quad (1)$$

where A_t is the architecture, W_t is the workload or task distribution, M_t is the persistent artifact state (code, plans, documentation, logs), and H_t is the available human-oversight budget. The architecture is a directed role graph

$$A_t = (R_t, E_t, B_t), \quad (2)$$

with roles R_t , communication edges E_t , and role-specific context or bandwidth budgets B_t .

Each round yields a vector of observables

$$y_t = (q_t, v_t, c_t, h_t) \in \mathbb{R}^4, \quad (3)$$

where q_t denotes quality or correctness, v_t delivery velocity, c_t coordination cost, and h_t human attention consumed. The specific metrics may differ by deployment, but the important point is that performance is *multi-objective*. No single scalar captures system health without contestable weighting assumptions.

Let $\mathcal{D}$ index monitored dimensions and let $s_d(y_t)$ be the normalized slack of dimension d relative to an unacceptable region or target threshold. The active bottleneck at round t is

$$b_t = \arg \min_{d \in \mathcal{D}} s_d(y_t). \quad (4)$$

Suppose an intervention π_t changes the architecture from A_t to A_{t+1} . We say that *bottleneck migration* occurs when the targeted dimension improves by at least $\epsilon > 0$ but the active bottleneck changes:

$$s_{b_t}(y_{t+1}) - s_{b_t}(y_t) \geq \epsilon \quad \text{and} \quad b_{t+1} \neq b_t. \quad (5)$$

This definition makes the concept operational: migration is not merely any post hoc story about change, but a measurable shift in the most constrained dimension after a structural intervention.

TODO: Instantiate y_t and $s_d(\cdot)$ for the present experiments. At minimum, the camera-ready version should specify how quality, velocity, coordination cost, and human effort are measured from logs, commits, tests, and review events. Without this measurement protocol, bottleneck labels remain partly interpretive.

5.2 Role Fission as a Capacity-Interference Tradeoff

The role-fission hypothesis can be stated as a comparison between overloaded local cognition and added coordination cost. For a role $r \in R_t$ with responsibility set U_r , define its effective cognitive load as

$$L_r(t) = \sum_{u \in U_r} \ell(u, t) + \sum_{\substack{u, v \in U_r \\ u \neq v}} \iota(u, v, t), \quad (6)$$

where $\ell(u, t)$ is the context demand of responsibility u and $\iota(u, v, t)$ captures interference between responsibilities that compete for the same context window or working memory. Let $B_r(t)$ denote the role's usable context budget and let $\Gamma(r_1, r_2, t)$ denote the communication overhead introduced by splitting r into two roles r_1 and r_2 .

Fission is predicted to be beneficial when the overload penalty exceeds the induced coordination cost:

$$L_r(t) - B_r(t) > \lambda \Gamma(r_1, r_2, t), \quad (7)$$

for some deployment-dependent conversion factor $\lambda > 0$. This formalization is intentionally simple, but it makes several qualitative predictions clear.

First, increasing context budget should delay the onset of fission. Second, improving summarization or memory artifacts should reduce the interference term $\iota(\cdot, \cdot, t)$ and can therefore substitute for some role splits. Third, as communication bandwidth degrades, the right-hand side grows and the net benefit of additional specialization should eventually reverse. These are experimentally testable claims, not merely interpretive analogies.

TODO: Test the inequality above with controlled interventions. Hold substrate and workload fixed while varying context budget, communication bandwidth, and the presence of summary-producing roles. The current Wallfacer evidence is observational and does not identify the causal contribution of each factor.

5.3 Trust Calibration as Preference Learning with Escalation

The autonomy boundary between an agentic system and its human overseers can be modeled as a context-dependent abstention problem. Let $\mathcal{X}$ denote a decision-context space derived from system state:

$$x_t = \phi(S_t, a_t), \quad (8)$$

where a_t is a candidate action and ϕ extracts relevant features such as task novelty, architectural configuration, recent failure history, and expected blast radius. Define a latent function

$$f : \mathcal{X} \rightarrow \mathbb{R}, \quad (9)$$

where $f(x)$ is the expected advantage of autonomous execution over human consultation in context x . We place a Gaussian Process prior

$$f \sim \mathcal{GP}(\mu_0, k), \quad (10)$$

with conservative prior mean μ_0 and kernel k .

As in standard preference learning [14, 32], we assume observations arrive as pairwise comparisons rather than calibrated scalar rewards:

$$z_{ij} = \begin{cases} 1 & \text{if outcome at } x_i \text{ is preferred to outcome at } x_j, \\ 0 & \text{otherwise.} \end{cases} \quad (11)$$

Using a probit likelihood,

$$P(z_{ij} = 1 \mid f) = \Phi\left(\frac{f(x_i) - f(x_j)}{\sqrt{2}\sigma}\right), \quad (12)$$

the posterior mean and uncertainty induce an escalation policy

$$\pi_{\text{auto}}(x) = \mathbf{1}[\mu_t(x) - \beta\sigma_t(x) > \tau], \quad (13)$$

where β controls risk aversion and τ is the autonomy threshold. The system proceeds autonomously only when the lower confidence bound exceeds the threshold; otherwise it escalates.

This formulation is close in spirit to reject-option learning [7]: the key question is not only which action is better, but when to defer. It also aligns with contextual dueling-bandit formulations [19], where policy improvement is driven by pairwise preference feedback under changing context.

TODO: Instantiate the feature map ϕ , kernel k , and escalation parameters (β, τ) on at least one real deployment. A scientifically stronger version of this paper should compare the learned policy against static review rules under a matched human-attention budget.

5.4 Architecture Search as Constrained Preferential Bayesian Optimization

If architecture is allowed to vary, the system faces a design problem over configurations $c \in \mathcal{C}$, where c encodes role count, topology, memory structure, and resource allocation. Let $u(c, w)$ denote latent utility under workload w , and let $g_j(c, w) \leq 0$ encode operational

constraints such as latency, token cost, test-failure rate, or human-review load. Because utility is multi-dimensional and difficult to scalarize honestly, we observe preferences of the form

$$(c_i, w) > (c_j, w) \quad (14)$$

instead of direct rewards. The natural optimization problem is therefore a constrained version of preferential Bayesian optimization [10, 26].

Conceptually, an Architect agent maintains a posterior over both utility and feasibility, then proposes the next architecture by balancing exploitation, exploration, and operational safety. This interpretation clarifies what the Architect is and is not. It is not a magical meta-agent that “knows” the right topology; it is a sequential experimental design mechanism operating over an architecture space with noisy, preference-based feedback.

TODO: The present paper does not contain such an optimization loop; it only motivates one. A critical next step is to run an offline replay or online intervention study in which candidate architectures are compared under a fixed workload distribution and a published preference protocol.

5.5 Falsifiable Implications

The formalization above yields four concrete predictions.

- (1) For a fixed workload, increasing effective context budget should delay or suppress role fission.
- (2) Summary-producing artifacts should partially substitute for organizational differentiation by reducing interference within a role.
- (3) Net performance as a function of role count should be non-monotone: early specialization helps, but coordination cost eventually dominates.
- (4) A learned uncertainty-aware escalation boundary should outperform static HITL rules at the same human-review budget when task contexts are heterogeneous.

These predictions are the main reason to formalize the framework at all. Without them, “architectural morphogenesis” would remain a vivid description; with them, it becomes a family of claims that can be contradicted by data.

6 THEORETICAL EXTENSIONS

The framework developed thus far rests on a single case study (Section 2), a fission principle (Section 3), and a formal construction (Section 4). In this section, we extend the framework through seven additional epochs, each introducing qualitatively new dynamics. We present these extensions not as findings but as a theoretical scaffold for future empirical investigation. The progression is organized so that each epoch’s characteristic bottleneck motivates the next.

TODO: For camera-ready clarity, consider adding a compact table that lists each speculative phase, its predicted observable signature, and the minimum experiment that would count as supporting or disconfirming evidence.

6.1 Epoch II: Population (Phases 9–10)

Phase 8’s self-reconfiguring organism begins to reproduce. In **Phase 9**, multiple system instances run in parallel with heterogeneous architectural configurations. A routing mechanism directs incoming

tasks to the best-matched instance, and the Architect gains organic preference pairs by comparing outcomes across configurations. This parallels the transition from individual learning to population-level selection in evolutionary biology [30]. Boyd and Richerson’s [9] dual-inheritance theory (genetic and cultural) suggests an analogous model for agentic systems: architectural configuration as the “genetic” channel and accumulated knowledge as the “cultural” channel.

The Ralph experiment (Section 3) provides an early empirical data point for this phase: three instances started from empty repositories and converged on the same problem domain (cellular automata) yet diverged into distinct specializations (scientific analysis, simulation breadth, and algorithmic diversity), with the degree of divergence modulated by a single prompt-level variable (the explore/exploit signal). This convergence-then-divergence pattern is precisely the population-level dynamic predicted here, observed in miniature before any inter-instance communication or routing mechanism was introduced.

In **Phase 10**, instances begin sharing knowledge through what we term a Knowledge Commons, allowing learning to propagate across the population without requiring each instance to rediscover the same patterns. Henrich’s [28] demographic model of cultural evolution predicts that larger populations maintain more complex cultural knowledge; if this analogy holds, agent populations with shared knowledge should exhibit qualitative capability gains relative to isolated instances. However, shared knowledge introduces the risk of *knowledge pollution*: an erroneous belief propagated from one instance infects others, analogous to misinformation cascades in social networks [57]. The binding constraint shifts from individual capability to collective epistemics.

6.2 Epoch III: Ecosystem (Phases 11–12)

In **Phase 11**, instances specialize into expert niches, and comparative advantage [46] emerges naturally from the population dynamics. Instances that happen to perform well on certain task types receive more of those tasks, reinforcing their specialization. This is the reverse of Conway’s Law [17]: it is not organizational structure that constrains architecture, but capability pressure that drives differentiation.

Phase 12 represents a critical transition: the human shifts from an external oracle, consulted across a static boundary, to a node within the system’s own topology. The human’s cognitive contribution (judgment, preference, domain knowledge) is routed through the same protocol layer as inter-agent communication. This does not diminish the human’s authority but does change its character: the human becomes one information source among several, distinguished by the quality of its signal rather than by structural privilege. This transition connects to ongoing debates about the role of human oversight in increasingly autonomous systems [23].

6.3 Epoch IV: Symbiosis (Phases 13–14)

In **Phase 13**, the HITL boundary transitions from a static rule (“always consult before deployment”) to a learned Trust GP (Section 4.1). The PBO mathematics developed earlier applies directly: the system learns which decision contexts warrant autonomous operation and which require human consultation, updating its trust

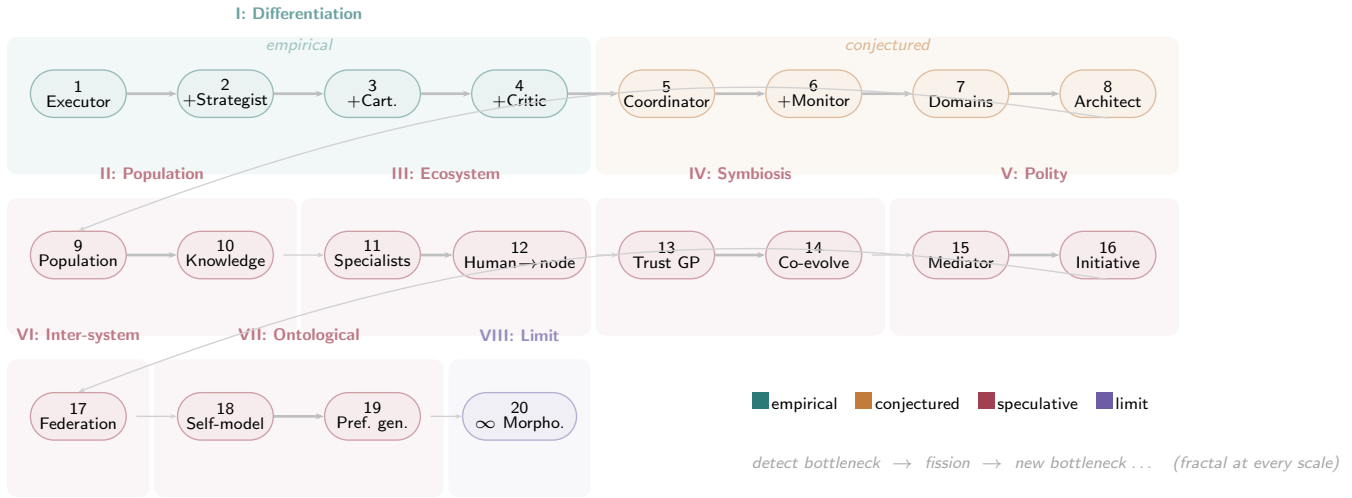


Figure 8: Roadmap of the framework’s progression across eight epochs and twenty phases, with explicit epistemic grading. Teal phases (1–4) were directly observed in the Wallfacer experiment. Orange phases (5–8) are theoretically conjectured based on the fission principle. Red phases (9–19) represent speculative extensions to populations, multi-stakeholder settings, inter-system federations, and self-referential dynamics. The purple terminal phase (20) represents the limiting claim that the stable attractor is morphogenesis itself. Each transition is driven by bottleneck migration.

model from observed preference pairs. Coordination cost, which rose monotonically through Epochs I–III, begins to decline. This mirrors the pattern in human organizations: initial role differentiation drives coordination costs upward, but once processes and trust are established, costs recede as coordination shifts from ad-hoc negotiation to learned automatic routing.

Phase 14 is the endpoint of the individual-system arc, if there is one. System and human principals co-evolve: the system’s outputs shape what its principals consider possible, while the principals’ evolving preferences shape the system’s behavior. This mirrors niche construction in evolutionary biology [43], in which organisms modify their environment, altering the selection pressures that act on subsequent generations. Neither party converges to a static equilibrium; the dynamic relationship is itself the stable attractor. The system has become not a tool but a collaborator, and the intelligence resides not in either party but in the loop between them.

6.4 Epoch V: Polity (Phases 15–16)

In realistic deployment scenarios, multiple human stakeholders with potentially conflicting preferences interact with the system simultaneously. In **Phase 15**, a *Mediator* role emerges. Arrow’s impossibility theorem [5] establishes that no preference aggregation function simultaneously satisfies a set of desirable axioms (unrestricted domain, Pareto efficiency, independence of irrelevant alternatives, and non-dictatorship). The Mediator therefore does not attempt to aggregate preferences into a single optimum but instead maintains and presents a Pareto front over the multi-dimensional outcome space [20], allowing human principals to navigate the tradeoff structure themselves.

Phase 16 is a qualitative leap: the system begins to propose changes that no human principal has requested. It detects cross-domain patterns invisible to any individual stakeholder and generates unsolicited proposals. This is no longer tool behavior but collegial behavior, and it immediately triggers legitimacy questions. Who authorized the proposal? Who is accountable if it fails? The transition from tool to collaborator, already implicit in Phase 14, becomes explicit and contested. We raise this possibility not to endorse it but to identify it as a critical frontier for both technical and normative investigation [23].

6.5 Epoch VI: Federation (Phase 17)

In **Phase 17**, multiple independent agentic ecosystems begin to interact. Each ecosystem has its own architectural configuration, knowledge base, and trust model. Inter-system communication requires a shared protocol layer; MCP [4] provides a communication primitive at this level. However, transmitting trust is an order of magnitude harder than transmitting data: a preference judgment calibrated within one ecosystem may not transfer to another with different operational history. The binding constraint is no longer internal coordination but inter-system trust negotiation.

6.6 Epoch VII: Ontological (Phases 18–19)

In **Phase 18**, the system’s self-model becomes a first-class object: the system begins to model its own blind spots. This is where Russell’s paradox and Lawvere’s fixed-point theorem [38] become concrete rather than analogical. A self-model that includes a representation of its own limitations is precisely the kind of self-referential structure that Lawvere’s theorem guarantees will have fixed points and that Gödel’s theorem [24] guarantees will be incomplete.

Phase 19 is the furthest extrapolation of the PBO framework: the system no longer merely discovers preferences but *generates*

them. The GP posterior over preference space permits extrapolation into unobserved regions, enabling the system to propose preference hypotheses that its human principals have not yet articulated. Both acceptance and rejection of such proposals constitute informative signal for posterior updates. This idea resonates with Dewey’s [18] pragmatist account of preferences as constructed through inquiry, and with Sen’s [50] critique of revealed preference theory. Whether such preference generation is technically feasible, epistemically legitimate, or socially desirable remains entirely open.

6.7 Epoch VIII: The Limit (Phase 20)

Phase 20 is not a specific architecture but a recognition: there is no final form. The entire progression from Phase 1 to Phase 20 is self-similar at every scale. Within a single agent role, the detect-bottleneck-fission-emerge cycle operates on context window resources. Across a pipeline, it operates on role differentiation. Across a population, it operates on configuration diversity. At the meta-architectural level, it operates on the capacity for self-modification itself. The pattern is fractal.

The stable attractor of this system is not a topology; it is the sustained capacity for topological change. Bottleneck migration is not a defect to be engineered away but the system’s heartbeat: the signal that the system remains far from equilibrium, that it is still alive in the thermodynamic sense of Prigogine and Nicolis [44]. If bottlenecks cease to migrate, the system has reached equilibrium, and equilibrium, in this context, is death.

7 ONTOLOGICAL LIMITS

This section is deliberately interpretive. It explores what the empirical and formal parts of the paper *might* imply if their central pattern generalizes, but it does not claim that the arguments below are established by the current data.

7.1 Morphogenesis as the Stable Attractor

Reviewing the progression from Phases 1 through the speculative extensions, we arrive at what we consider the paper’s central theoretical claim. If the pattern of bottleneck migration is genuine (an empirical question), and if it persists at higher levels of architectural complexity (a conjecture), then the system’s long-term trajectory is not toward any particular fixed architecture but toward a sustained capacity for architectural change. The cycle of detecting a bottleneck, fissioning a role, and observing the emergence of a new bottleneck would repeat at every scale: within individual agent roles, across pipelines, across populations, and at the meta-architectural level.

This characterization aligns with Prigogine and Nicolis’ [44] theory of dissipative structures: systems maintained far from thermodynamic equilibrium through continuous flows of energy or information. The agentic ecosystem, on this view, persists not despite its continuous reorganization but because of it. If bottlenecks cease to migrate, the system has reached equilibrium, which in this context would indicate stagnation. The connection to Friston’s [22] Free Energy Principle is also suggestive: the Architect’s self-model functions as a generative model of the system, bottleneck detection corresponds to surprise minimization, and role fission corresponds to active inference. We stress, however, that these are analogies, not

demonstrations; whether the mathematical structure of dissipative systems or free energy minimization actually applies to agentic architectures is a question for future formal analysis.

TODO: Either supply a minimal formal argument for the dissipative-structure analogy or narrow the claim further. At present this subsection is intellectually suggestive but evidentially weak relative to the rest of the paper.

7.2 Self-Reference and Incompleteness

If a system constructs an explicit model of its own capabilities and limitations (as we hypothesize for Phase 8 and beyond), it enters a self-referential regime that can be analyzed using two classical results. Lawvere’s fixed-point theorem [38] guarantees that in any cartesian closed category with a point-surjective morphism $\varphi : A \rightarrow B^A$, every endomorphism $f : B \rightarrow B$ has a fixed point. A verifier that verifies itself is precisely such a fixed point. Lawvere’s theorem establishes that such fixed points exist in any sufficiently rich formal setting; the question is not whether self-referential equilibria are possible but what properties they have. Gödel’s first incompleteness theorem [24] constrains the answer: any self-model powerful enough to be useful is necessarily incomplete. The system can know that it has blind spots but cannot enumerate all of them from within.

Taken together, these results suggest that self-reconfiguring agentic systems can achieve stable but irreducibly partial self-understanding. This connects to Hofstadter’s [29] theory of strange loops, Yanofsky’s [62] unified treatment of self-referential paradoxes, and Barwise and Moss’s [8] non-well-founded set theory. Von Foerster’s [56] second-order cybernetics arrives at the same structure from the operational side: when an observer recursively observes its own observations, the process converges to stable *eigen-behaviors*: fixed points that the system can reliably maintain, not because they have been externally certified but because they are self-consistent solutions of the recursive process. Lawvere’s theorem guarantees the existence of such fixed points in any sufficiently rich formal setting; Von Foerster’s framework explains how they are realized in practice through recursive interaction rather than top-down computation. We note that applying these mathematical results to engineered software systems involves a degree of analogical reasoning that may or may not survive formal scrutiny; we present them as a conceptual framework, not as a proof.

TODO: Clarify the mapping between the categorical objects in Lawvere’s theorem and concrete components of an agentic system, or else present the theorem strictly as an analogy. Reviewers will likely ask where the formal correspondence starts and stops.

7.3 Ontological Pluralism and the Observer

Finally, we note three meta-level observations that bear on the framework’s interpretation. First, a single system event (e.g., a role fission) can be described as “role differentiation” in organizational vocabulary, “symmetry breaking” in thermodynamic vocabulary [2], “strategy differentiation” in game-theoretic vocabulary [54], or “niche divergence” in ecological vocabulary [33]. Each description language reveals structures that the others occlude; a system described in only one vocabulary may exhibit systematic blind spots undetectable from within that vocabulary.

Second, if all components of a system are replaceable (today’s LLM agent could be tomorrow’s human engineer), the system’s identity resides not in its substrate but in its relational structure: its information flow topology, preference constraints, and coordination protocols. This is the category-theoretic perspective on identity [37], and it suggests that agentic system design should prioritize protocol-level interoperability over implementation-level homogeneity.

Third, the act of describing an agentic system is itself a component of the system it describes: a researcher analyzing the Wallfacer experiment functions as a Strategist (generating hypotheses), the analysis serves as an Executor (producing formalized descriptions), and the resulting paper acts as a Cartographer (creating artifacts that inform future experimental design). This self-referential observation, anticipated by Maturana and Varela’s [41] concept of autopoiesis, does not invalidate the framework but does caution against treating any description of an evolving system as final.

8 RELATED WORK

The paper sits at the intersection of four literatures: LLM-based agent systems, self-adaptive software architectures, preference-based oversight and optimization, and older organizational or cybernetic theories of regulation.

8.1 LLM Agents and Agentic Software Engineering

Recent LLM-agent work has established a broad design space for role decomposition, memory, reflection, and tool use. Single-agent systems such as Reflexion [52] and Voyager [58] emphasize verbal self-improvement or open-ended skill acquisition. Multi-agent systems such as CAMEL [39], AutoGen [60], MetaGPT [31], ChatDev [45], and Magentic-One [21] study role assignment, communication protocols, and orchestration patterns. In software engineering specifically, SWE-agent [61] demonstrates that interface design between the agent and the repository environment is a first-order determinant of performance.

Our work differs from this literature in two respects. First, the dependent variable is not benchmark success under a fixed architecture, but endogenous architectural change under sustained operation. Second, the empirical setting is open-ended rather than fully task-specified: Ralph instances begin from empty repositories with no seed task, and Wallfacer is allowed to adapt its own role structure over time. The closest overlap is therefore not with benchmark-driven comparisons alone, but with work on open-ended agent trajectories and long-horizon systems behavior.

8.2 Self-Adaptive and Autonomic Software Systems

The most direct systems lineage is autonomic and self-adaptive software. Kephart and Chess’s vision of autonomic computing [34] and architecture-based self-adaptation frameworks such as Rainbow [13] treat monitoring, diagnosis, planning, and execution as explicit feedback loops over a software architecture model. Brun et al. [12] frame self-adaptive systems as systems that modify behavior in response to internal and external change, while preserving higher-level goals.

The present paper can be read as an agentic extension of that tradition. The proposed Cartographer resembles an evolving architecture knowledge model; the conjectured Coordinator and Monitor resemble specialized control-loop components; and the Architect role generalizes a MAPE-K style controller to a setting in which the decision-making elements are themselves LLM agents that can emit proposals, summaries, and preference judgments. The main difference is that the adaptive substrate is no longer a conventional controller but an organization of stochastic language agents with limited context windows and high communication costs.

8.3 Preference Learning, Abstention, and Human Oversight

The formal core of the paper builds on Gaussian Process preference learning [14], active preference learning [32], and preferential Bayesian optimization [26]. These methods replace absolute reward values with ordinal comparisons, which is attractive when stakeholders can state which outcome they prefer but cannot assign reliable scalar scores. In online settings, the same family of ideas connects to contextual dueling bandits [19], where the system learns from pairwise comparisons conditioned on context.

Our Trust GP also relates to the literature on *abstention* or *rejection* learning [7]. The practical question is not only which action is best, but when the system should decline autonomy and escalate to a human. Relative to prior work, the contribution here is to treat the human-oversight boundary itself as a learned, context-dependent decision surface over architectural state, workload, and uncertainty, rather than as a static hand-authored rule.

8.4 Organizational, Cybernetic, and Security Antecedents

The role-fission argument draws on organizational design and coordination theory. Simon’s near-decomposability [53], Mintzberg’s structural configurations [42], and Malone and Crowston’s coordination theory [40] all explain why differentiation can improve local performance while creating new coordination burdens. Brooks’ communication-cost argument [11] and Conway’s Law [17] provide adjacent intuitions about when additional structure helps and when it becomes a liability.

The bottleneck-migration perspective is also shaped by cybernetics and security. Ashby’s Law of Requisite Variety [6] and the Good Regulator theorem [16] motivate the idea that stronger regulation requires richer internal models, while Thompson’s “trusting trust” attack [55], Trusted Computing Base reasoning [48], Byzantine agreement [35], and Zero Trust Architecture [47] show how verification and trust allocation remain structural problems even in engineered systems. These works do not directly study LLM-agent organizations, but they supply the conceptual language for understanding why verification pressures can produce architectural differentiation rather than disappear.

Finally, the paper’s more speculative sections draw on second-order cybernetics, autopoiesis, and self-reference [24, 38, 41, 56]. We keep those connections, but treat them as conceptual extensions rather than part of the paper’s empirical core. That distinction matters: the scientific contribution of the present draft is stronger

when these ideas are positioned as hypotheses and interpretive lenses, not as already-validated explanations.

9 DISCUSSION

9.1 Practical Implications

For practitioners building autonomous agentic systems, the framework suggests several design heuristics, which we offer tentatively pending broader empirical validation. First, it may be advisable to begin with the simplest architecture that functions and to add roles only when bottleneck signals are clear and persistent; premature fission incurs coordination overhead without commensurate capability gain. Second, tracking the *location* of bottlenecks over time may provide a more informative diagnostic of system health than monitoring any single performance metric. Third, preference comparison rather than scalar metrics may offer a more tractable and honest approach to architecture evaluation, given the multi-dimensional nature of system health. Fourth, designing for substrate independence (communicating via protocols rather than shared implementation details) enables component substitution, whether replacing one LLM with another or replacing an agent with a human, without disrupting organizational architecture. Fifth, the HITL boundary should be learnable rather than static, as human cognitive bandwidth is typically the scarcest resource in agentic deployments.

9.2 Threats to Validity

Construct validity. Several central quantities, especially “bottleneck,” “coordination cost,” and “self-awareness,” are currently inferred from qualitative evidence more than from preregistered operational definitions. The formalization in Section 5 reduces this ambiguity, but the current experiments do not yet instantiate all of the proposed measurements.

Internal validity. Wallfacer’s architectural transitions were adaptive interventions chosen in response to observed failures. That is appropriate for exploratory systems research, but it means the paper cannot cleanly attribute each downstream change to a single manipulated variable. Ralph offers a stronger comparison, yet its 2+1 design remains underpowered for decisive causal claims about the explore/exploit signal.

External validity. All runs use the same LLM substrate and stay within software-engineering tasks. The convergence on cellular automata may reflect a genuine attractor in the model’s prior over open-ended projects, but it may also be a training-distribution artifact. Likewise, the approximate line-count thresholds observed in Wallfacer should not be treated as universal phase transitions.

Theoretical validity. The connections to Lawvere’s theorem, Gödel’s incompleteness, dissipative structures, and autopoiesis are intended as interpretive and hypothesis-generating, not as deductions from the data. The paper is strongest when these sections are read as conceptual boundary-marking rather than as validated mechanism.

9.3 Experimental Agenda

The shortest path from the current draft to a stronger scientific paper is a focused experimental program rather than more metaphor.

Four pieces of evidence would substantially raise confidence in the framework.

- (1) Multi-seed replications across at least two model substrates, so that convergence and divergence can be separated from substrate-specific priors.
- (2) Controlled fission ablations that vary role count, context budget, and communication bandwidth while holding workload fixed.
- (3) A direct evaluation of the Trust GP against static human-escalation policies under a fixed review budget.
- (4) Publication of prompts, logs, intervention criteria, and annotation protocol so that other groups can reproduce the same trajectory analysis.

TODO: If space permits, add a compact “threats to validity” table and a “minimum evidence for camera-ready” table. These two additions would make the scientific status of each claim immediately legible to reviewers.

Several substantive questions remain open even after such a program. Can a normalized complexity metric predict fission thresholds across task domains? Do long-running agentic systems converge, oscillate, or drift? Under what conditions does role differentiation improve net utility rather than merely redistribute failure modes? How should independent agentic ecosystems negotiate trust, protocol compatibility, and preference exchange once interaction crosses system boundaries? These are now clearer questions than they were at the start of the paper, which is itself progress.

10 CONCLUSION

We have presented an empirical systems study, a formal scaffold, and a research agenda for architectural change in autonomous agentic systems. The strongest claims are modest but useful: in the observed runs, bottlenecks did not disappear, they migrated; independent runs did not wander arbitrarily, they converged on a task family and then diverged in specialization; and role differentiation appears to trade local cognitive relief against global coordination cost.

The paper is intentionally explicit about what remains missing. The role-fission principle is a testable hypothesis, not an established law. The Trust GP is a formal proposal, not yet a validated controller. The later sections on populations, self-reference, and morphogenesis as a stable attractor are conjectural extensions that help organize future experiments but currently outrun the data. If subsequent work reproduces the empirical regularities, operationalizes the proposed measurements, and validates the uncertainty-aware escalation model, then the field will have a stronger basis for treating architecture itself as a learnable variable in autonomous agent systems. If those tests fail, the value of the present paper will still be that it states clearly what should be measured next.

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